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RESEARCH

Genetic Variability of Transpiration Response to Vapor Pressure Deficit among Soybean Cultivars

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ABSTRACT

Simulation studies have demonstrated that the existence of a limitation on maximum transpiration rate (TR) at high air vapor pressure deficit (VPD) in water-limited conditions could result in significant yield increases. A genotype of soybean [*Glycine max* (L.) Merr.] (PI 416937) has been identified both with a slow-wilting phenotype in the field and limited TR above a VPD breakpoint (BP) of 2 kPa. Until now, there has been no full documentation of any other soybean genotype that has a restriction on TR at high VPD. The objective of this study was to extend the observations on TR response to VPD to a wider genetic base. Seven soybean genotypes were studied: three cultivars of broad genetic background and four new lines of diverse backgrounds with PI 416937 in their pedigrees. Data collected across a VPD range of ~1 to ~3.5 kPa for each genotype showed a two-segment TR response to VPD for two cultivars, neither of which was derived from PI 416937. Since three progeny of PI 416937 expressed slow-wilting in the field, these results indicate that the slow-wilting phenotype may also result from a mechanism other than the TR response to VPD. Identification of two new genetic selections with limitation of TR at high VPD indicates there is more than one genetic source for this potentially beneficial trait for water-limited environments.

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Abbreviations: BP, breakpoint in the TR response to VPD; PPFD, photosynthetic photon flux density; TR, transpiration rate; VPD, vapor pressure deficit.

A limitation on transpiration rate (TR) at high levels of atmospheric vapor pressure deficit (VPD) is a plant mechanism that could be exploited to allow soil water conservation and to allow sustained crop growth during dry periods in the later part of the growing season. In a model analysis of sorghum (*Sorghum bicolor* L.) growth, Sinclair et al. (2005) showed that a hypothetical cultivar in which TR was limited to a maximum rate under high VPD would result in yield increases in four locations in Australia in approximately 75% of the growing seasons. We (T. Sinclair, C. Messina, and C. Loeffler, personal communication) have done a similar analysis for soybean [*Glycine max* (L.) Merr.] in the United States. In this study, a hypothetical soybean cultivar with no further increase in TR above a VPD of 2 kPa resulted in yield increases in about 65% of the years in most production areas of the United States. It appears likely there would be substantial value in identifying and combining germplasm with limited TR at high VPD to develop commercial soybean cultivars.

Some crops, such as rice [(*Oryza sativa* L.), Morison and Gifford, 1983], grapevine [(*Vitis vinifera* L.), Patakas et al., 2005], and orange [(*Citrus sinensis* L. Osbeck), Oguntunde et al., 2007], have shown limited TR at high VPD. Other studies indicated intraspecific variability in this trait by comparing different cultivars of a given crop such as banana [(*Musa* sp), Thomas et

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al., 1998], cotton [(*Gossypium hirsutum* L.), Wang et al., 2004], and winter wheat [(*Triticum aestivum* L.), Xue et al., 2004]. For example, in the latter study, two cultivars had continually increasing TR as VPD was increased, while two other cultivars displayed a decreasing TR above a VPD of 3 kPa. Bunce (1984) compared the TR in three soybean cultivars and found that leaf conductance at 3 kPa was less than at 1 kPa. A difficulty in these previous studies, however, is that the TR response was measured either indirectly through sap flow measurements (e.g., Patakas et al., 2005; Oguntunde et al., 2007; Wang et al., 2004) or on single leaves (e.g., Morison and Gifford, 1983; Bunce, 1984; Thomas et al., 1998; Xue et al., 2004). Indeed, it has been reported that under some conditions such as high VPD, sap flow can be temporarily higher or lower than actual TR (Jarvis, 1975; Katerji et al., 1986; Alarcón et al., 2000) and that the hydraulic response of an isolated leaf could be quite different from the integrated response of an entire plant (Rawson et al., 1977; Johnson and Ferrell, 1983).

There have been recent studies to document the TR response to VPD of whole soybean plants among genotypes. Fletcher et al. (2007) developed a relatively simple experimental system in which whole plant TR could be measured while subjecting plants to a wide range of VPD conditions. Twelve plants were studied at one time. Results on the basis of a study of four soybean genotypes showed genetic variability in that a genotype identified in the field as "slow-wilting" (PI 416937) had a two-segment TR response to VPD. Above a breakpoint (BP) of about 2 kPa, TR was essentially constant. The other three cultivars studied by Fletcher et al. (2007) showed little or no change in the increase in TR across the studied range of VPD. Sinclair et al. (2008) examined in greater detail the unique response of PI 416937 to increasing VPD. They confirmed that there was a BP in PI 416937 at about 2 kPa and that the limitation on TR at higher VPD was linked to low hydraulic conductance at the leaf level, which was not observed in two genotypes that had no BP in the TR response to VPD.

Since the genetic background of soybean in which the two-segment TR response to VPD has been limited to genotype PI 416937, the objective of this study was to initiate exploration of the existence of this trait in a more diverse range of germplasm. Seven genotypes of diverse genetic backgrounds were included in this study: four lines that have PI 416937 in their pedigree and three lines that do not. The genotypes derived from PI 416937 were selected to gain some insight about how readily the TR response of PI 416937 to VPD might be transferred to productive progeny lines. The system developed by Fletcher et al. (2007) was used to measure TR on whole plants over a range of VPD.

MATERIALS AND METHODS

Genetic Material and Experiments

Four new cultivar lines developed by T. Carter (ARS-USDA, Raleigh, NC) were included in this study, all belonging to Maturity Group VII and having the slow-wilting genotype PI 416937 in their pedigree (Table 1). The contribution of PI 416937 to each genotype varied. One genotype, N7001, was observed to express normal wilting during developing soil water deficit while the other three (N96-7031, N01-11771, N01-11136) were observed to exhibit slow wilting. Three additional cultivars were included in the study that did not have PI 416937 in their pedigree (Table 1). Among these cultivars was Davis, a historic U.S. cultivar first released in 1966 (Salado-Navarro et al., 1993) that is normal wilting. A second cultivar in this group, NTCPR94-5157, was derived from a cross using Davis. This second cultivar has been observed to be slow wilting. A third cultivar was derived from a cross of Tracy and Ransom, N73-1102, and its wilting characteristic has yet to be documented.

The tests of TR response to VPD were conducted in a glasshouse in Gainesville, FL (29°38′ N, 82°22′ W). In total, six experiments were performed to document for each genotype the TR response to VPD (Table 2). Each genotype was studied in two independent experiments, where the number of replicate plants for a genotype was either three or four. Obtaining results from two experiments for each genotype ensured that data were collected over the target range of VPD levels and allowed a verification of consistency in response across experiments. Combining results of the two experiments, data were collected from seven to eight replicate plants for each genotype.

Experimental Measurements

The measurement of TR response to VPD was done by using the experimental system developed by Fletcher et al. (2007). Seeds were sown in polyvinyl chloride columns (0.1-m diameter and 0.2-m tall) filled with 1.5 to 2 kg of compost garden soil (Miracle-Gro Lawn Products, Inc., Marysville, OH) containing slow-release fertilizer (15% N–5% P–10% K). A small hole was drilled in the center of the end cap of the pot for drainage. A PVC toilet flange had been glued to the top open end of the pots to facilitate attachment of the VPD chamber.

Three seeds inoculated with *Bradyrhyzobium japonicum* (Nitragin, Inc., Brookfield, WI) were sown in each pot. Eight to 15 d after sowing, each pot was thinned to a single plant. Pots were watered every 1 to 2 d. Plants were grown for 3 to 6 wk until the projected canopy diameters of the plants were near the diameter of the chamber in which the TR measurements were made (0.3 m). Due to genotypic differences in canopy architecture, the size criterion was used for determining measurement dates rather than a specific phenological stage in development.

The day before TR measurements, plants were placed in a second glasshouse and the pots were watered to dripping. A 0.34-m diameter lid of a food container (Rubbermaid Commercial Products LLC, Winchester, VA) with the center cut out was slipped over the plants and attached to the toilet flange. Aluminum foil was placed over the soil and sealed around the stem of the plant to eliminate soil evaporation into the chamber. The following morning the aerial parts of each plant were sealed into a 0.3-m tall, 21-L transparent plastic food container (Rubbermaid

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Commercial Products LLC, Winchester, VA). The inverted container was placed over the plant and attached to the previously installed lid. Each container was equipped with a 12 V, 76-mm diameter computer box fan (Northern Tool and Equipment, Burnsville, MN) to mix the air inside the chamber. A humidity/temperature sensor (Extech Instruments, League City, TX) was mounted through the side wall of each container.

In general, light levels inside the greenhouse were approximately 70% of the irradiance incident to the greenhouse. In addition, the TR containers extinguished approximately 25% of the solar radiation (Fletcher et al., 2007). Therefore on cloudless days, the light levels in the containers could have been as high as about 1000 μmol m^{-2} s^{-1}. To avoid excessive temperatures (>35°C) in the containers on cloudless days, a partial shade was erected in the greenhouse. The highest photosynthetic photon flux density (PPFD) values, measured external to the TR chambers, typically reached ~720 μmol m^{-2} s^{-1} on average (±200 SD); thus the PPFD exposure to the plants was approximately 500 μmol m^{-2} s^{-1}.

For each experiment, measurements were performed on two consecutive days. Three VPD levels were targeted (0 to 1.5 kPa, 1.5 to 2.5 kPa, 2.5 to 3.5 kPa) on both days. These treatments were applied sequentially, from the lowest to the highest. The main source of VPD variation was achieved by flowing different volumetric flow and different air sources into the containers to balance the humidifying effect of the transpiring leaves (Fletcher et al., 2007). The highest humidity (lowest VPD) was achieved by flowing air at 5 L min^{-1} using two pumps (DOA0P704, Gast, Benton Harbor, MI) with their intake tubes inside an atomizing humidifier (Hermidifier, Sanford, NC). All flow rates were monitored using flow meters (Model FL-2043, Omega, Stamford, CT), one for each chamber. Medium humidity was achieved by flowing air into the containers at a rate of 25 to 30 L min^{-1} using two 57-L air compressors (Kobalt model L-215902, Lowe's Companies Inc., North Wilkesboro, NC). The lowest humidity was achieved by flowing air from the two compressors through a PVC tube (0.10-m diameter, 0.38-m length) filled with a silica gel desiccant (6–16 mesh, Grade 49; S160-500, Fisher Scientific, Pittsburgh, PA) at a rate of about 30 L min^{-1}.

Each humidity treatment was imposed for about 60 min. Before each treatment, the humidity was set and the container unit with the plant inside was allowed to equilibrate for 20 to 30 min. After equilibration, the entire unit of plant, container, and pot were weighed on a balance with a resolution of 0.1 g (Model SI-8001, Denver Instrument, Denver, CO). The time of the initiation of the measurement period was recorded. Temperature and relative humidity were recorded three to six times during the approximately 60-min measurement period. At the end of the measurement period, the time was recorded and the weighing procedure was repeated so that the rate of mass change attributed to TR could be calculated. The corresponding VPD values were calculated from the temperature and relative humidity data and averaged for the measurement period. On the second day, the entire sequence of measurements was repeated.

Following TR measurements on the second day, leaf area of each plant was determined so the TR data for plants could be normalized relative to leaf area. Since there was shading within each plant canopy, these calculations represent an approximation at normalizing TR to account for some of the size differences that existed among plants. Leaves were removed from the plants and passed through an area meter (Model LI-3100, Licor, Lincoln,

Table 1. Genetic material used in the study.

Genotype	Maturity group	Descriptor†	Genetic background		Reference§
			Parentage‡	PI 416937 in pedigree	
				[Y/N]	
N7001	VII	NW	N77-114 × PI 416937	Y	*b,c*
N96-7031	VII	SW	N7001 × N90-7241	Y	*a,c*
N01-11771	VII	SW	GRAHAM × N96-7031	Y	*a,c*
N01-11136	VII	SW	NTCPR94-5157 × N96-7031	Y	*a,c*
Davis	VI	NW	D49-2573 × N45-1497	N	*a,c*
N73-1102	Unknown	Unknown	Tracy × Ransom¶	N	*a,c*
NTCPR94-5157	VI	SW	Davis × N73-1102	N	*a,c*

†Wilting properties observed in the field (T. Carter, personal communication): NW, normal wilting; SW, slow wilting.

‡Female parent × male parent.

§Reference: *a*, Gillen and Shelton (2006); *b*, Carter et al. (2003); *c*, T. Carter, personal communication.

¶Both parents are normal wilting.

Table 2. Summary of the experiments.

Experiment†	Genotypes	Replicates	Sowing date	Measurement date‡	VPD§ range	
					Min.	Max.
					— kPa —	
E1	Davis | N01-11136 | N01-11771	4	1/11/2007	13 & 14/12/2007	0.83	3.30
E2	N73-1102	4	9/1/2008	13 & 14/2/2008	0.68	3.46
E3	N73-1102	4	21/2/2008	2 & 3/4/2008	0.87	3.50
E4	N01-11136 | N01-11771	3	2/6/2008	23 & 24/6/2008	1.46	3.15
E5	NTCPR94-5157 | N7001 | N96-7031 | Davis	3	6/6/2008	30/6 & 1/7/2008	1.64	3.28
E6	NTCPR94-5157 | N7001 | N96-7031	4	2/7/2008	23 & 24/7/2008	1.01	3.51

†Experiments are presented in chronological order.

‡Whole-plant leaf areas and aerial dry weights were measured on the second day of each experiment.

§VPD, vapor pressure deficit.

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NE). Leaves and the remaining aerial parts of each plant were placed in a drying oven for 3 to 6 d to determine dry weight.

Data and Statistical Analysis

Before statistical analysis, the data were examined using a program developed in the R language (R Development Core Team, 2008). A simple loop-based program was developed to plot the data of individual replicates of each genotype and visually identify irrelevant measurements. These data resulted from errors in weighing, plants with leaves injured by the fan, or leaf damage observed as yellowing or pest injuries. Data filters were also included in the program to exclude data where the container temperature exceeded 35°C.

The filtered data were then submitted to a segmented linear regression analysis for each genotype using GraphPad Prism (GraphPad Software Inc., San Diego, CA). This software was based on a user-defined iterative function, developed by Motulsky (1999, p. 194) and previously used by Fletcher et al. (2007). The algorithm seeks to minimize the sum of squares by independently fitting linear regressions to two segments of data. The criterion for convergence of the iteration was when the sum of squares of successive iterations differed by less than 0.01%. The two-segment model was assumed if the slopes of the two linear regressions were significantly different ($p < 0.05$); otherwise all the data were fitted to a single linear regression.

Statistical analyses were performed with the R-package "stats" (R Development Core Team, 2008). Analysis of variance, covariance, and linear model fittings were performed by the "*lm*" function, which is an adaptation of the S-language function of the same name developed by Chambers (1992). Two-sample *t* test analyses were performed by the function of the same name in the "stats" package. Shapiro-Wilk normality test and Pearson correlation analysis were performed respectively through the functions "*shapiro.test*" and "*cor*."

Table 3. Plant size data and segmented regression results for the two genotypes found to have a breakpoint.

		Genotype	
		N73-1102	NTCPR94-5157
Plant size			
Leaf area (cm^2)	Mean	608.6	688.7
	SD†	186.9	131.3
Dry weight (g)	Mean	2.73	3.48
	SD	0.96	0.45
Segmented Regression			
Breakpoint (kPa)	Value	1.94	2.17
	SE‡	0.16	0.19
Slope #1 (mg H_2O m^{-2} s^{-1} kPa^{-1})	Value	22.92	31.31
	SE	3.31	4.36
	n§	17	16
Slope #2 (mg H_2O m^{-2} s^{-1} kPa^{-1})	Value	2.70	9.51
	SE	2.4	4.24
	n	26	24
Overall r^2		0.78	0.85

†Standard deviation of leaf area and dry weight values measured on seven to eight replicate plants for each genotype.

‡Standard errors on the segmented regression parameters.

§Number of values for each regression segment.

RESULTS

For all experiments and genotypes, temperatures were normally distributed (Shapiro-Wilk normality test), with an average at 32.0°C (±2.8 SD). The main basis for differences in VPD among treatments was the humidity of the air, resulting in VPD measurements ranging from ~1 to ~3.5 kPa for all genotypes. In all cases, the target range of VPD was achieved within each of the two experiments, so that replicating the experiments served as a test for experimental repeatability.

As anticipated, plant size varied among the seven or eight replicate plants of each genotype. The mean plant mass for each genotype ranged from 2.50 g (±0.31 SD) to 3.48 g (±0.48 SD). An analysis of variance of dry weights among the cultivars showed a significant genotypic effect ($p < 0.05$). Mean leaf area among the genotypes ranged from 570.6 (±76.8 SD) to 703.5 cm^2 (±196 SD) (Tables 3 and 4). Leaf area was not significantly different among the genotypes.

For all lines, no significant differences were found within genotypes in the regression results for the TR response to VPD when the data collected during the two experiments for each genotype were analyzed independently. Hence, the data collected in the two experiments for each genotype were combined. Major differences in the TR response to VPD were obtained among these genotypes. The two-segment regression analysis identified two genotypes with significant BP (Fig. 1b and 1d, Table 3), and the other five genotypes (Fig. 1a and 1c, Table 4) had no significant BP. The BP for N73-1102 was at 1.94 kPa (±0.16 SE) and for NTCPR94-5157 at 2.17 kPa (±0.19 SE). Genotype NTCPR94-5157 has been observed to have the slow-wilting phenotype, and the phenotype of N73-1102 has yet to be documented in the field. Curiously, none of the four genotypes with slow-wilting PI 416937 in their pedigree exhibited a BP in the TR response to VPD.

The two genotypes with a BP differed ($p < 0.01$) in their slopes of TR as a function of VPD at VPD below the BP. Genotype N73-1102 had a slope of 22.9 mg H_2O m^{-2} s^{-1} kPa^{-1} at low VPD, while NTCPR94-5157 had a slope of 31.3 mg H_2O m^{-2} s^{-1} kPa^{-1} at low VPD. These divergent slopes at low VPD corresponded, however, with the overall range of slopes found in the other five genotypes. Four of these other genotypes had overall slopes in the range of 20.2 to 23.3 mg H_2O m^{-2} s^{-1} kPa^{-1} corresponding with the initial slope of N73-1102, while one line (N01-11136)

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Table 4. Plant size data and linear regression results for the five genotypes fitted by a single linear regression.

		Genotype				
		N7001	N96-7031	Davis	N01-11771	N01-11136
Plant size						
Leaf area (cm^2)	Mean	675.4	600.4	703.5	570.6	659.8
	SD†	112.0	155.9	196.0	76.8	90.4
Dry weight (g)	Mean	3.34	3.24	3.04	2.50	2.89
	SD	0.37	0.68	0.59	0.31	0.26
Linear Regression						
Slope (mg H_2O m^{-2} s^{-1} kPa^{-1})	Value	20.17	20.74	22.84	23.30	29.73
	SE‡	2.23	2.03	2.26	1.52	2.01
Intercept (mg H_2O m^{-2} s^{-1})	Value	–10.47	–10.64	–18.37	–12.51	–27.40
	SE	5.23	4.97	5.41	3.29	4.37
Total number of values (n)		40	38	39	33	41
r^2		0.68	0.74	0.73	0.88	0.85

†Standard deviation of leaf area and dry weight values measured on seven to eight replicate plants for each genotype.

‡Standard errors on the regression parameters.

had an overall slope of 29.7 mg H_2O m^{-2} s^{-1} kPa^{-1}, corresponding with the initial slope of NTCPR94-5157 (Table 3). The highest and lowest slopes were significantly different (at $p < 0.05$), and this variability was found not to be associated with either leaf area or shoot dry weights (Fig. 2a and 2b, Tables 3 and 4).

Of course, the major distinguishing difference among the genotypes was the low slope in the plot of TR vs. VPD at VPD higher than the BP for genotypes N73-1102 and NTCPR94-5157. The slope at the higher VPD was only 2.7 mg H_2O m^{-2} s^{-1} kPa^{-1} (±2.4 SE) for N73-1102 and 9.5 mg H_2O m^{-2} s^{-1} kPa^{-1} (±4.24 SE) for NTCPR94-5157, which are both substantially less than the overall linear slope in the remaining five cultivars (Table 4).

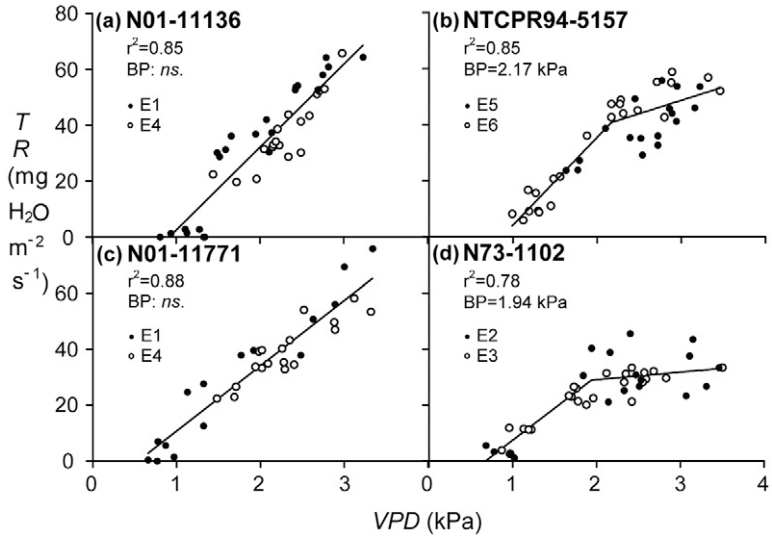


Figure 1. Transpiration rate (TR) response to vapor pressure deficit (VPD) of four soybean [*Glycine max* (L.) Merr.] genotypes. For each genotype, the two symbols identify data collected in the two independent experiments (see Table 1 for details). Segmented and linear regressions fittings were applied according to the approach detailed in the methods section; BP, breakpoint in the TR response to VPD.

DISCUSSION

The present study confirmed genetic variability in whole plant TR response to VPD (i.e., a single linear regression vs. a two-segment regression) in soybean, first found by Fletcher et al. (2007). These results, however, showed that the two-segment TR response to VPD was not unique to PI 416937, since two additional cultivars (N73-1102 and NTCPR94-5157) were identified as expressing a BP in the TR response to VPD. It was rather surprising that the two genotypes displaying the two-segment TR response to VPD were not derived from PI 416937. The expression of the BP response in these two additional genotypes, however, was very similar to PI 416937 with BP values of approximately 2 kPa. These results showed that the phenotype of limited TR at high VPD exists in more than one germplasm resource.

On the other hand, the four genotypes derived from PI 416937 did not express restricted

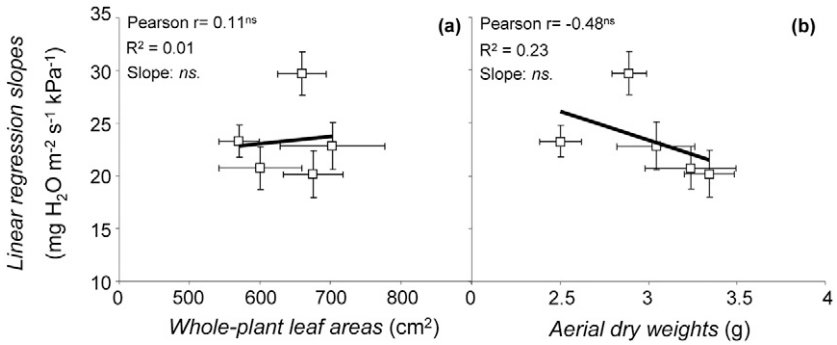


Figure 2. Correlations between the slopes of the linear transpiration rate (TR) response to vapor pressure deficit (VPD) observed for (a) five lines and their corresponding leaf areas and (b) dry weights. Error bars for both TR and VPD are indicated for each datum. Pearson correlation and R^2 values are given; ns, nonsignificant.

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TR at high VPD. These limited results indicate that this trait may not be readily transferred to progeny lines. The simplest explanation is that this trait might be recessive. Another nonindependent possibility is that the phenotype of limited TR at high VPD is under a complex, multigenic control, so that trait expression in progeny lines is variable. Since three of these lines have been phenotyped as exhibiting slow wilting in the field, these results indicate that the two-segment response of TR to VPD is not the only basis for expression of the slow-wilting phenotype. This conclusion is consistent with the observations of Fletcher et al. (2007) in that the slow-wilting genotype PI 471938 did not exhibit a sensitivity of TR to VPD.

The slope of the regression of TR against VPD reflects the gaseous conductance of water vapor diffusion from these plants. Since the water storage capacitance in these plants relative to the 60-min test period is negligible, and the slopes were independent from leaf areas and aerial dry weights, these regression results seem to reflect a basic limitation on water transfer. Results of the slope at low VPD for the two genotypes with a BP and the other five genotypes segregated into two groups. The two genotypes expressing a BP were divided between the two groups of differing TR slope at low VPD. It seems that the basis of the BP response is not associated with the TR potential at low VPD. The group of five genotypes had slopes approximately 85% of the slope of the other two genotypes with a BP. Possible interpretations of these results are that either stomata conductance or the hydraulic conductance within the plant for these five genotypes may be smaller at low VPD than the other two genotypes. More intensive investigation is required to resolve the basis for these differences in water loss at low VPD.

The low slope of N73-1102 and NTCPR94-5157 at high VPD indicates that a hydraulic limitation exists in these genotypes that restricts water flow at high VPD. The study of Sinclair et al. (2008) showed that there was a limitation of hydraulic flow in the leaves of PI 416937 associated with its expression of limited TR above the BP. It is unknown whether the same mechanism exists in these two additional genotypes, but the existence of a BP at approximately the same VPD and the same near-zero slopes at high VPD indicates that the basis of the response may be similar.

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References

Alarcón, J.J., R. Domingo, S. Green, M.J. Sánchez-Blanco, P. Rodríguez, and A. Torrecillas. 2000. Sap flow as an indicator of transpiration and the water status of young apricot trees. Plant Soil 227:77–85.

Bunce, J.A. 1984. Identifying soybean lines differing in gas exchange sensitivity to humidity. Ann. Appl. Biol. 105:313–318.

Carter, T.E., Jr., J.W. Burton, D.T. Bowman, Z. Cui, X. Zhou, M.R. Villagarcia, M.O. Fountain, and A.S. Niewoehner. 2003. Registration of 'N7001' soybean. Crop Sci. 43:1126–1127.

Chambers, J.M. 1992. Linear models. p. 95–138. *In* J.M. Chambers and T.J. Hastie (ed.) Statistical models in S. Wadsworth & Brooks/Cole Pacific Grove.

Fletcher, A.L., T.R. Sinclair, and L.H. Allen, Jr. 2007. Transpiration responses to vapor pressure deficit in well watered "slow-wilting" and commercial soybean. Environ. Exp. Bot. 61:145–151.

Gillen, A.M., and G.W. Shelton (ed.). 2006. Uniform soybean tests report—southern states 2006. USDA–ARS, Crop Genetics and Production Research Unit, Stoneville, MS.

Jarvis, P.G. 1975. Water transfer in plants. p. 369–394. *In* D.A. de Vries and N.K. van Alfen (ed.) Heat and mass transfer in the plant environment. Part 1. Scripta Book, Washington, DC.

Johnson, J.D., and W.K. Ferrell. 1983. Stomatal response to vapour pressure deficit and the effect of plant water stress. Plant Cell Environ. 6:451–456.

Katerji, N., M. Hallaire, Y. Menoux-Boyer, and B. Durand. 1986. Modeling diurnal patterns of leaf water potential in field conditions. Ecol. Model. 33:185–203.

Morison, J.I.L., and R.M. Gifford. 1983. Stomatal sensitivity to carbon dioxide and humidity: A comparison of two C_3 and two C_4 grass species. Plant Physiol. 71:789–796.

Motulsky, H.J. 1999. Analyzing data with GraphPad Prism, GraphPad Software, San Diego CA.

Oguntunde, P.G., N. van de Giesen, and H.H.G. Savenije. 2007. Measurement and modelling of transpiration of a rain-fed citrus orchard under subhumid tropical conditions. Agric. Water Manage. 89:200–208.

Patakas, A., B. Noitsakis, and A. Chouzouri. 2005. Optimization of irrigation water use in grapevines using the relationship between transpiration and plant water status. Agric. Ecosyst. Environ. 106:253–259.

R Development Core Team. 2008. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.

Rawson, H.M., J.E. Begg, and R.G. Woodward. 1977. The effect of atmospheric humidity on photosynthesis, transpiration and water use efficiency of leaves of several plant species. Planta 134:5–10.

Salado-Navarro, L.R., T.R. Sinclair, and K. Hinson. 1993. Changes in yield and seed growth traits in soybean cultivars released in the southern USA from 1945 to 1983. Crop Sci. 33:1204–1209.

Sinclair, T.R., G.L. Hammer, and E.J. van Oosterom. 2005. Potential yield and water-use efficiency benefits in sorghum from limited maximum transpiration rate. Funct. Plant Biol. 32:945–952.

Sinclair, T.R., M.A. Zwieniecki, and N.M. Holbrook. 2008. Low leaf hydraulic conductance associated with drought tolerance in soybean. Physiol. Plant. 132:446–451.

Thomas, D.S., D.W. Turner, and D. Eamus. 1998. Independent effects of the environment on the leaf gas exchange of three banana (*Musa* sp.) cultivars of different genomic constitution. Sci. Hortic. (Amsterdam) 75:41–57.

Wang, C., A. Isoda, Z. Li, and P. Wang. 2004. Transpiration and leaf movement of cotton cultivars grown in the field under arid conditions. Plant Prod. Sci. 7:266–270.

Xue, Q., A. Weiss, T.J. Arkebauer, and P.S. Baenziger. 2004. Influence of soil water status and atmospheric vapor pressure deficit on leaf gas exchange in field-grown winter wheat. Environ. Exp. Bot. 51:167–179.

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